

HW 4

$$1. \quad i(t) = \frac{1}{R} V_s(t) + i_L$$

$$V_s = L \frac{d\dot{i}_L(t)}{dt}$$

$$i_L(t) = \frac{1}{L} \int_{-\infty}^t V_s(t) dt$$

$$i(t) = \frac{1}{R} V_s(t) + \frac{1}{L} \int_{-\infty}^t V_s(t) dt$$

$$h(t) = \frac{1}{R} \delta(t) + \frac{1}{L} u(t)$$

$$i_R(t) = \frac{e^{-\lambda t} u(t)}{R}$$

$$i_L(t) = \frac{1}{L} \int_{-\infty}^t e^{-\lambda t} u(t) dt = \frac{1}{L} \left. \frac{1}{-\lambda} e^{-\lambda t} \right|_0^t dt$$

$$= \frac{1}{L\lambda} (1 - e^{-\lambda t}) u(t)$$

$$i(t) = \frac{e^{-\lambda t} u(t)}{R} + \frac{1}{L\lambda} (1 - e^{-\lambda t}) u(t)$$

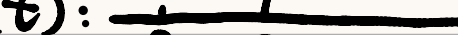
2. $t \in (1, 4)$ $x(t) \neq 0$

$$t \in (-1, 1) \quad h(t) \neq 0$$

$$\therefore \begin{cases} t+1 > 1 \\ t-1 < 4 \end{cases} \Rightarrow t \in (0, 5)$$

$$\therefore \text{ when } t \in (0, 5) \quad y(t) = x(t) * h(t) \neq 0$$

3. (a) $x(t)$:  $h(t)$: 

$h(-T)$:  $h(t-T)$: 

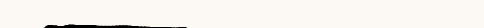
$h(t-T) = -e^{-\frac{t-T}{2}} u(t-T-2)$

① $-2+t < 2 \Rightarrow t < 4$ $y(t)=0$

② $-2+t > 2 \Rightarrow t > 4$

$$y(t) = \int_2^{t-2} -e^{\frac{\tau}{2} - \frac{t}{2}} d\tau = -2e^{\frac{\tau}{2} - \frac{t}{2}} \Big|_2^{t-2} = -2e^{-1} + 2e^{\frac{t}{2}}$$

$$y(t) = -2e^{-t} (1 - e^{-\frac{t}{2} + 2}) u(t-4)$$

b. $x(t)$:  $h(t)$: 

$h(t-\tau)$:  $e^{x-\tau} u(1-t+\tau)$

$$\textcircled{1} \quad t-1 < 0 \Rightarrow t < 1 \quad y(t) = \int_0^2 e^{2t-2\tau} d\tau - \int_2^5 e^{2t-2\tau} d\tau = e^{2t-4} + \frac{1}{2}e^{2t-2} - \frac{1}{2}e^{2t-10}$$

$$\textcircled{2} \quad t \in (1, 3) \quad y(t) = \int_{t-1}^2 e^{2t-\tau} d\tau - \int_2^3 e^{2t-\tau} d\tau = e^{2t-4} + \frac{1}{2}e^2 + \frac{1}{2}e^{2t-10}$$

③ $t \in (3, 6)$ $y(t) = -\frac{1}{2} \int_{t-1}^3 e^{2t-2\tau} d\tau = -\frac{1}{2} e^{2t} + \frac{1}{2} e^{2t-6}$

④ $t > 6, y(t) = 0$

C. $x(t)$:  $h(t)$: 

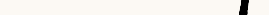
$h(t-T)$: 

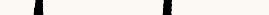
① $t < 1$ $y(t) = 0$

$$\textcircled{2} t \in (1, 3) \quad y(t) = \int_0^{t-1} 2 \sin \pi t \, dt = -\frac{2}{\pi} \cos \pi t \Big|_0^{t-1} = -\frac{2}{\pi} [\cos \pi(t-1) - 1]$$

③ $t \in (3, 5)$ $y(t) = \int_{t-3}^2 2 \sin \pi \tau = -\frac{2}{\pi} \cos \pi \tau \Big|_{t-3}^2 = -\frac{2}{\pi} [1 - \cos(\pi t - 3\pi)]$

④ $t > 5$ $y(t) = 0$

d. $x(t)$:  $h(t)$: 

$h(t-T)$:  $e^{-t+T} \sin(t-T)$

① $t < 0$ $y(t) = 0$

$$\textcircled{2} t \in (0, 1) \quad y(t) = \int_0^t e^{-t+\tau} \sin(t-\tau) d\tau = -\frac{1}{2} e^{-t} (\sin t + \cos t) \Big|_0^t$$

$$= -\frac{1}{2} e^{-t} (\sin t + \cos t) + \frac{1}{2}$$

$$\textcircled{3} t > 1 \quad y(t) = \int_0^1 e^{-t+\tau} \sin(t-\tau) d\tau = \left. \frac{1}{2} e^{-t+\tau} (\sin \tau + \cos \tau) \right|_0^1$$
$$= \frac{1}{2} e^{-t+1} \sin(t-1) + \cos(t-1) - \frac{1}{2} e^{-t} (\sin t + \cos t)$$

4. $h''(t) + 2h'(t) + h(t) = \delta'(t) + 2\delta(t)$

$$h_r(t) = (C_1 + C_2 t) e^{-t} \quad h(t) = h_r(t) u(t)$$

$$h'(t) = h_r'(t) u(t) + h_r(t) \delta(t)$$

$$h''(t) = h_r''(t) u(t) + h_r'(t) \delta(t) + h_r(t) \delta'(t)$$

$$h''(t) + 2h'(t) + h(t) = [h_r''(t) + 2h_r'(t) + h_r(t)]u(t) + [2h_r(t) + h_r'(t)]\delta(t) + h_r(t)\delta'(t)$$

$$\therefore h_r(0)=1 \quad h_r'(t)=0$$

$$h_r(t) = (C_1 + C_2 t) e^{-t} \quad h_r(0) = C_1 = 1$$

$$h_r'(t) = C_2 e^{-t} - (1 + C_2 t) e^{-t} = 0 \Rightarrow C_2 = 1$$

$$\therefore h(t) = (1+t)e^{-t} u(t)$$

$$b. \quad h'''(t) + 4h''(t) + 5h'(t) + 2h(t) = \delta''(t) + 3\delta(t)$$

$$r^3 + 4r^2 + 5r + 2 = 0 \Rightarrow r^3 + 2r^2 + r + 2r^2 + 4r + 2$$

$$= (r+1)^2(r+2)$$

$$h(t) = [(C_1 + C_2 t)e^{-t} + C_3 e^{-2t}] u(t)$$

$$h(t) = h_r(t) u(t)$$

$$h'(t) = h_r'(t) u(t) + h_r(0) \delta(t)$$

$$h''(t) = h''_r(t) u(t) + h_r'(0) \delta(t) + h_r(0) \delta'(t)$$

$$h'''(t) = h'''_r(t) u(t) + h''_r(0) \delta(t) + h_r'(0) \delta'(t) + h_r(0) \delta''(t)$$

$$\begin{cases} h''(0) + 4h_r'(0) + 5h_r(0) = 3 \\ h_r'(0) + 4h_r(0) = 0 \\ h_r(0) = 1 \end{cases} \Rightarrow \begin{cases} h_r(0) = 1 \\ h_r'(0) = -4 \\ h_r''(0) = 14 \end{cases} \quad -11$$

$$h_r(0) = C_1 + C_3 = 1$$

$$h_r'(0) = C_2 - C_1 - 2C_3 = -4 \Rightarrow \begin{cases} C_1 = -6 \\ C_2 = 4 \\ C_3 = 7 \end{cases}$$

$$h_r''(0) = C_1 - 2C_2 + 4C_3 = 14$$

$$\therefore h(t) = (-6 + 4t)e^{-t} + 7e^{-2t}$$